

PROGRAM 8: Create kubernetes pods using yaml description

PROJECT STRUCTURE

>nodejs-configmap.yaml

>nodejs-pod.yaml

>[server.js](#)

Step 1: nano nodejs-configmap.yaml

```
GNU nano 7.2
apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
    const http = require("http");
    const port = 3000;

    const server = http.createServer((req, res) => {
      res.end("Hello from Node.js running inside Kubernetes!");
    });

    server.listen(port, () => {
      console.log(`Server running at port ${port}`);
    });
```

Step2: nano configmap.yaml

```
GNU nano 7.2                                     nodejs-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: nodejs-pod
  labels:
    app: nodejs
spec:
  containers:
    - name: nodejs
      image: node:18
      command: ["node", "/usr/src/app/server.js"]
      ports:
        - containerPort: 3000
      volumeMounts:
        - name: app-code
          mountPath: /usr/src/app
  volumes:
    - name: app-code
      configMap:
        name: nodejs-app-config
```

Step3: nano sever.js

```
GNU nano 7.2                                     server.js
const http = require("http");
const port = 3000;

const server = http.createServer((req, res) => {
  res.end("Hello from Node.js running inside Kubernetes!");
});

server.listen(port, () => {
  console.log(`Server running at port ${port}`);
});
```

step4:Apply the yaml files

>kubectl apply -f nodejs-configmap.yaml

>kubectl apply -f nodejs-pod.yaml

Step5: check status

Kubectl get pods

Step6: expose as service

>kubectl port-forward pod/nodejs-pod 3000:3000

```
Jan 5 12:52
Terminal
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~$ cd p8
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/p8$ ls
nodejs-configmap.yaml  nodejs-pod.yaml  server.js
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/p8$ nano nodejs-configmap.yaml
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/p8$ nano nodejs-pod.yaml
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/p8$ nano server.js
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/p8$ kubectl apply -f nodejs-configmap.yaml
kubectl apply -f nodejs-pod.yaml
configmap/nodejs-app-config created
pod/nodejs-pod created
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/p8$ kubectl get pods
NAME                                READY   STATUS              RESTARTS   AGE
nginx-deployment-6f9664446b-5j8rb   1/1     Running             3 (4d21h ago)  6d23h
nginx-deployment-6f9664446b-hzjrm   1/1     Running             3 (4d21h ago)  6d23h
nginx-deployment-6f9664446b-w9npc   1/1     Running             0           30m
nodejs-pod                           0/1     ContainerCreating   0           11s
1RV24MC104_SINDHURA@sindhra-IdeaPad-Slim-5-16IAH8:~/p8$ kubectl port-forward pod/nodejs-pod 3000:3000
```

